Pairs Trading on Sector ETFs: Method and Backtest Results

1 Data and setup

Daily dividend-adjusted closing prices of 11 sector ETFs (XLK, XLF, IYR, XLV, XLE, XLI, XLY, XLP, XLB, XLU, VOX), 2004-09-29 to 2026-10-08. All prices are used in logs. The data are split by time: the first 70% is **in-sample** (2004-09-29 to 2020-02-26, 3,879 days) and is the only data used to select and fit pairs; the last 30% is **out-of-sample** (2020-02-27 to 2026-10-08, 1,663 days). Nothing fitted in-sample is updated out-of-sample unless stated. Positions are decided at the close of day $t-1$ and earn the return of day t. Each unit of position change costs 5 bp. Returns are per \$1 of gross exposure.

2 Stationarity test (in-sample)

For every pair of ETFs (Y, X) (55 pairs), regress log prices on the in-sample data and form the spread

$$s_t = \ln P_t^Y - a - b \ln P_t^X, \qquad (a, b) = \text{OLS fit}. \tag{1}$$

The spread is tested with the Engle–Granger test (an ADF test on s_t with critical values for an estimated residual). Both orderings of each pair are tried and the lower p-value is kept. A pair is **stationary** if $p < 0.10$ (10% level). *Example.* For XLB / XLU the fit gives $b = 0.866$: the spread is the part of XLB not explained by XLU. A stationary spread keeps returning to its mean; a non-stationary one drifts away.

Table 1: Pairs that are stationary in-sample (9 of 55). σ is the in-sample standard deviation of the spread.

Pair Y/X	EG t-stat	p-value	b	a	σ
XLU / XLK	-4.19	0.004	0.708	0.644	0.081
XLU / XLI	-3.95	0.009	0.833	-0.415	0.086
XLB / XLU	-3.87	0.011	0.866	0.535	0.093
XLV / XLY	-3.58	0.026	0.891	0.860	0.080
XLB / XLP	-3.50	0.032	0.725	0.378	0.105
XLI / XLV	-3.47	0.035	0.865	0.386	0.087
XLU / XLY	-3.35	0.048	0.654	0.488	0.104
XLB / XLI	-3.22	0.067	0.753	0.064	0.068
XLB / XLV	-3.18	0.073	0.644	0.383	0.108

3 Static-mean strategy

For each stationary pair, the in-sample mean μ and standard deviation σ of the spread are frozen, and the z-score is

$$z_t = \frac{s_t - \mu}{\sigma}. \tag{2}$$

Rule: short the spread when $z_t \geq 1.5$ (short Y, long b of X), long the spread when $z_t \leq -1.5$, and close when $|z_t| \leq 0.25$. The position is set at the close and earns the next day's return. Daily P&L is the return of the spread position per \$1 of gross exposure, less a 5 bp cost on any change in position. *Example.* Short the spread with $b = 0.866$. If Y falls 1.0% and X rises 0.5% in a day, the spread falls by 1.43% ($1.0\% + 0.866 \times 0.5\%$), a gain of 0.77% per \$1 of gross exposure (1.43% divided by $1 + 0.866$). Out-of-sample, the frozen spread is tested again with a standard ADF test (Table 2); the portfolio is the equal-weight average of the pair strategies.

4 Rolling-mean strategy

Same rule, hedge ratio b and σ, but the frozen mean is replaced by the average of the spread over the last 3 years (the window ends at t), so $z_t = (s_t - \bar{s}_t)/\sigma$ with $\bar{s}_t$ the 3-year average. *Example.* If a spread drifts away and stays there, the rolling mean follows it and z_t returns toward zero; with a frozen mean, z_t stays large.

Table 2: Out-of-sample results by pair. ADF = test of the frozen out-of-sample spread. Return in %; Sharpe is annualised, no risk-free rate subtracted.

	Out-of-sample ADF test			Static		Rolling 3 years	
Pair	ADF stat	p	trades	Return	Sharpe	Return	Sharpe
XLU / XLK	-2.02	0.278	1	-29.3	-0.35	+17.9	0.30
XLU / XLI	-2.53	0.109	4	+24.2	0.54	+25.9	0.58
XLB / XLU	-3.32	0.014	5	+32.4	0.54	+25.0	0.52
XLV / XLY	-2.20	0.205	7	+50.8	0.86	+18.1	0.38
XLB / XLP	-3.92	0.002	2	-2.3	0.01	+14.3	0.36
XLI / XLV	-1.01	0.748	3	+3.0	0.10	-4.7	-0.07
XLU / XLY	-2.61	0.090	5	+42.4	0.65	+34.2	0.64
XLB / XLI	-2.22	0.198	3	+16.9	0.50	-2.5	-0.05
XLB / XLV	-2.77	0.063	3	+11.9	0.23	+20.2	0.44

5 Portfolios and SPY overlay

Sharpe ratio = annualised average daily return divided by the annualised standard deviation of daily returns; no risk-free rate is subtracted.

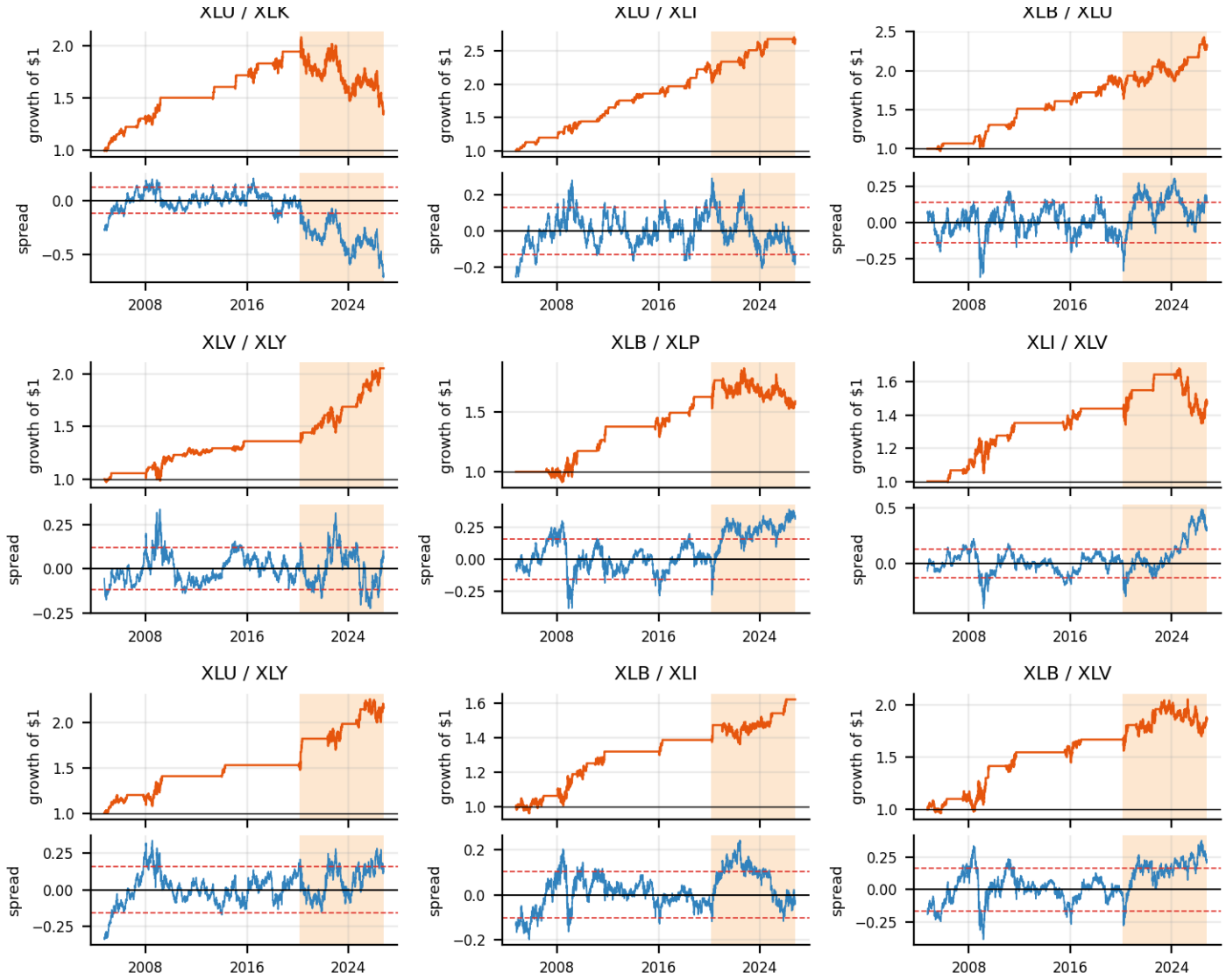


Figure 1: Static-mean strategy, one block per pair. Top: growth of \$1. Bottom: spread s_t (blue), frozen in-sample mean (black) and $\pm 1.5\sigma$ entry bands (red dashed). Orange shading = out-of-sample.

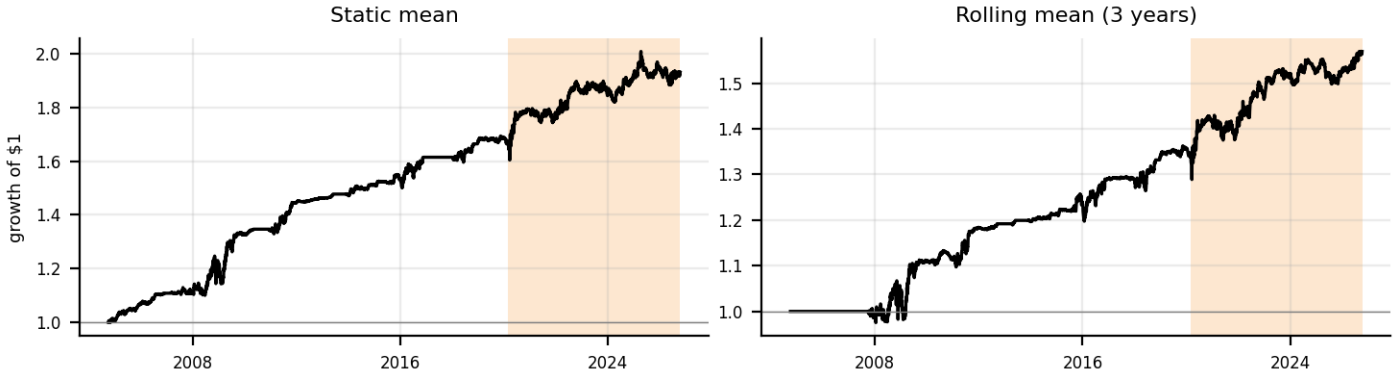


Figure 3: Combined portfolio of the 9 pair strategies, each with 1/9 of the capital: growth of \$1 for the static-mean and rolling-mean strategies.

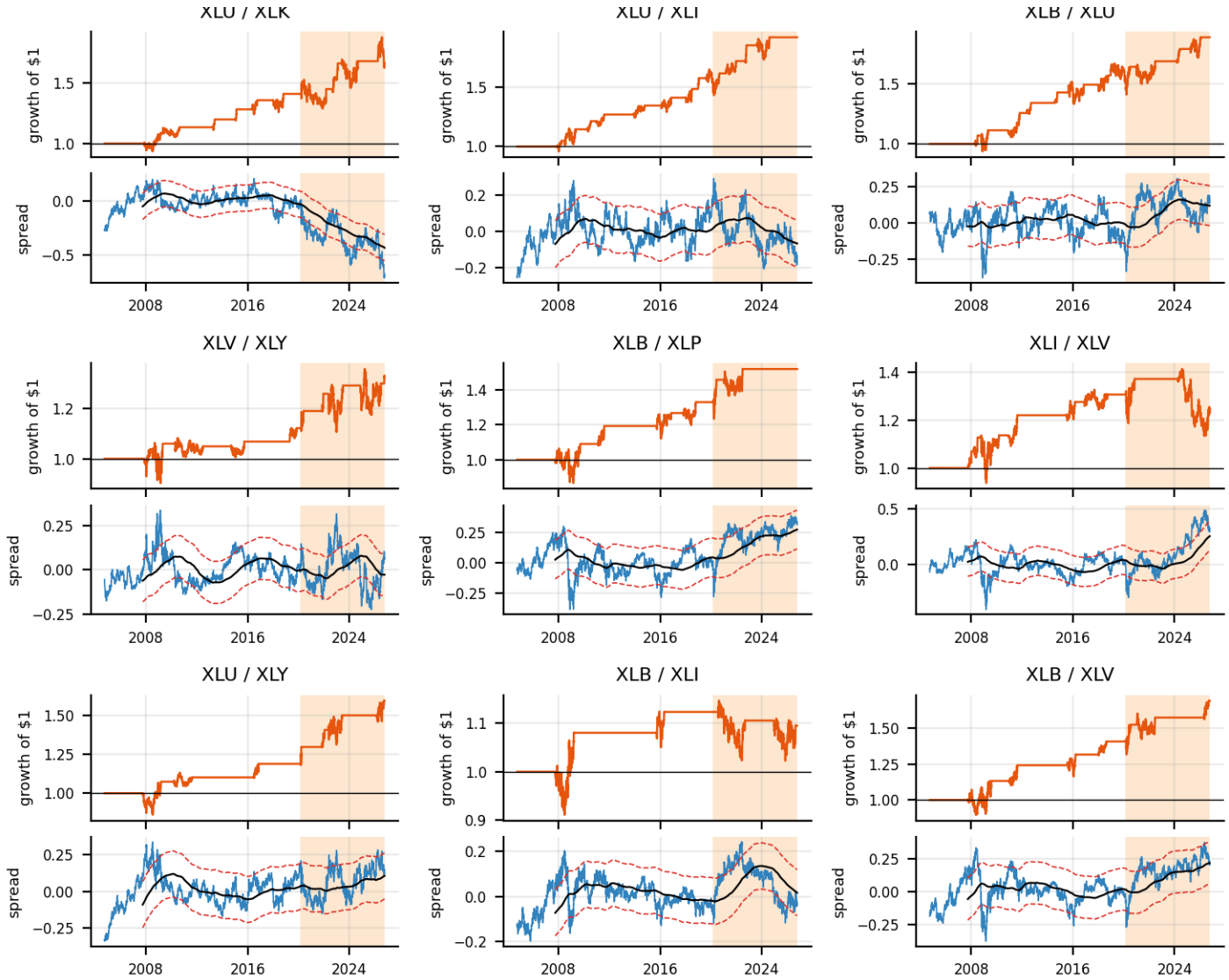


Figure 2: Rolling-mean strategy, one block per pair. Top: growth of $1. Bottom: spread (blue), 3-year rolling mean (black) and $\pm 1.5\sigma$ bands around it (red dashed).

Table 3: Portfolio backtests. Return and maximum drawdown in %. Last column: average share of capital in SPY, out-of-sample.

	In-sample			Out-of-sample			
Strategy	Return	Max DD	Sharpe	Return	Max DD	Sharpe	% in SPY
Static mean, idle cash earns 0	+66.2	-8.3	0.90	+16.3	-6.2	0.55	–
Rolling mean (3 years), idle cash earns 0	+33.9	-8.0	0.53	+17.3	-3.9	0.68	–
Pairs (static mean) + SPY	+458.1	-17.8	1.01	+55.0	-15.8	0.88	33
Pairs (rolling mean, 3 years) + SPY	+456.3	-19.4	0.98	+112.2	-14.6	1.07	58
SPY buy and hold	+279.1	-55.2	0.57	+173.2	-28.3	0.86	100

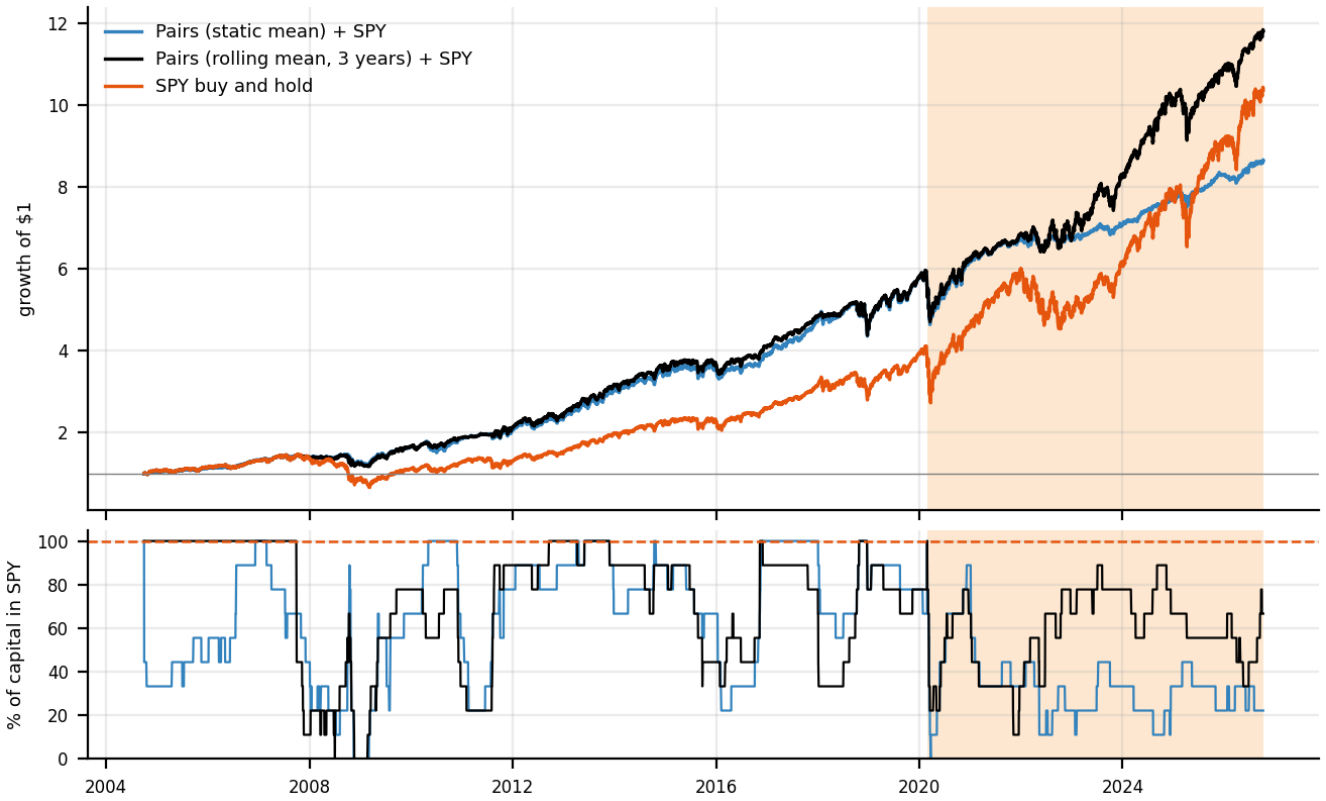


Figure 4: Growth of $1 (top) and share of capital held in SPY (bottom) for the two SPY-overlay strategies (static mean and rolling mean) and SPY buy-and-hold.